SOPHIA DA COSTA FERNANDES

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PROFESSIONAL OBJECTIVE

To work in backend development and software architecture, building efficient and scalable systems for real-world problems and contributing to the delivery of clean, well-structured code.

PROFESSIONAL SUMMARY

Software Engineering student at PUC Minas, focused on software architecture and backend development. 1 year and 3 months of corporate experience at Stellantis working with automotive embedded systems, ECU validation and cloud data integration. Track record of hands-on personal projects with Java, Spring Boot, IoT, APIs and mobile development, including an award for Best Practical Project of the first semester at PUC Minas. Full portfolio at sophiafernandes.dev. Fluent in English.

PROFESSIONAL EXPERIENCE

Stellantis | Engineering Intern | *June 2025 to September 2026 (1 year and 3 months)*

- Developed an embedded solution for collecting data from test benches and vehicles using Raspberry Pi + ESP32 + MCP2515, adopted by the C&I team in Electrical Engineering.
- Validated automotive systems with a focus on Infotainment (Radio and Cluster), including failure analysis and debugging of embedded systems to ensure the integrity of subsystem communication over the CAN network.
- Integrated telemetry flows with cloud infrastructure within the connected vehicles team, ensuring real-time data synchronization between the vehicle and external platforms.

PROJECTS

Cod Invest: Financial Asset Management Platform | *2026 (in progress)*

- Full-stack application for tracking and analyzing investment funds, integrated with the BrapiDev API to consume real financial data. Focused on end-user experience and clean code practices, built with Java, JavaScript, Bootstrap and Spring Boot.
- Repository: github.com/sophiafernandesc/cod-invest-project

Diagnosticar: IoT Vehicle Diagnostics System | *2025*

- Full-stack IoT system for real-time vehicle diagnostics: data pipeline with embedded CAN data collection (Arduino Nano + MCP2515), an ESP32 bridge, an MQTT broker for cloud infrastructure and a presentation layer in an Android app built with React Native.
- Repository: github.com/sophiafernandesc/esp32-canbus-obd2-project

Voltra: Fuel Management and Vehicle Tracking App | *2025*

- Android app built with Flutter for fuel management and vehicle tracking, reading fuel consumption parameters through an ELM327 scanner integrated with an ESP32 and a tracking module, with data persistence in MySQL.
- Repository: github.com/sophiafernandesc/voltra-project

Spindle: Quadruped Robot | *2024*

- Quadruped robot with 8 servo motors for articulation and locomotion, programmed in C++, with a mobile control app built in App Inventor. Awarded Best Practical Project of the first semester (2024/1) of Computer Engineering at PUC Minas.
- Repository: github.com/sophiafernandesc/spindle-project

More projects and details at sophiafernandes.dev.

EDUCATION

Bachelor's Degree in Software Engineering | PUC Minas | *2026 to present*

Focus on software architecture. Expected graduation: 2029.

Bachelor's Degree in Computer Engineering | PUC Minas | *2024 to 2026*

Attended until the first semester of 2026, then transferred to Software Engineering.

EXTRACURRICULAR ACTIVITIES

- **WebTech Extension Project** | PUC Minas (2026 to present).
- **Quantum Computing Study Group** | PUC Minas (second semester of 2024), focused on exploring the theoretical foundations and applications of the field.
- **Automotive Networks** | PUC Minas: technical study of embedded communication protocols (CAN, LIN and Automotive Ethernet), focused on systems integration and vehicle connectivity with cloud services.

SKILLS

Backend: Java, Spring Boot, REST APIs, OOP, Clean Code, MySQL

Frontend / Mobile: JavaScript, React, React Native, Flutter, Bootstrap

Embedded / IoT: C/C++, Arduino, ESP32, CAN Bus, MQTT

Cloud / DevOps: MQTT Broker, cloud infrastructure integration

Tools: Git, GitHub, Linux (Red Hat)

CERTIFICATES

Red Hat Linux Course: Fundamentals of Linux-based operating systems.

Java Course (Udemy): Programming logic, Object-Oriented Programming and software structuring.

LANGUAGES

Portuguese: Native

English: Fluent (reading, writing and speaking)